UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

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PART A (80 MARKS)

QUESTION 1

a) Using Caesar Cipher technique, decrypt the following ciphertext with a shift of 7 (replace each letter with the one 7 positions to the left).
KLCPJLYLABYULKHUKDPWLKVUL

(2 marks)

b) [THEORY PLACEHOLDER - classical ciphers, 6 marks, State. No sourced question passed verification.]

(6 marks)

QUESTION 2

a) Explain how a rail fence transposition cipher encrypts and decrypts a message.

(2 marks)

b) Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

(6 marks)

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Vigenere tableau (A = 0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

QUESTION 3

a) [COMPUTED PLACEHOLDER - columnar_3x4, 2 marks]

(2 marks)

b) Describe six properties and cryptanalytic characteristics of classical cipher families, covering substitution, Vigenere, Caesar shift and columnar transposition.

(6 marks)

QUESTION 4

a) State TWO (2) formal security models and the specific security concern each addresses.

(2 marks)

b) [COMPUTED PLACEHOLDER - acm, 6 marks]

(6 marks)

QUESTION 5

a) Explain how substitution ciphers hide messages and how the Arabs broke monoalphabetic substitution.

(2 marks)

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b) Decrypt the following ciphertext using columnar transposition with a 4 x 6 matrix (4 rows, 6 columns). Put the ciphertext into the columns, read the plaintext out of the rows, and ignore the filler X at the end. The key is the keyword secret.
FOIENCXLRPYXOMTSEEEUCLRX

(6 marks)

QUESTION 6

a) [THEORY PLACEHOLDER - access control models, 2 marks, List. No sourced question passed verification.]

(2 marks)

b) Discuss the goals that authorization mechanisms pursue and the role of reference monitors, then explain how the Chinese Wall model's simple security rule and its *-property each address a specific security concern.

(6 marks)

QUESTION 7

a) Describe the two stages of the biometric authentication process.

(2 marks)

b) Describe six points about biometric mechanisms and technologies, covering their definition, categories, and specific technologies with their limitations.

(6 marks)

QUESTION 8

a) Explain how the TPM anchors a chain of trust during platform boot.

(2 marks)

b) Refer to your answer in (a). Explain six components and functions of the NGSCB/Nexus architecture that support trusted computing on a platform.

(6 marks)

QUESTION 9

a) Explain two characteristics of Trojan horse malware that distinguish it from other malicious programs.

(2 marks)

b) Compare how a boot sector virus and a rootkit each achieve their malicious effect on an infected system, and explain why the rootkit is harder to detect.

(6 marks)

QUESTION 10

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a) State TWO (2) impacts of malware attacks on the confidentiality, integrity or availability of information.

(2 marks)

b) Explain how the Samy worm used cross-site scripting to propagate itself and describe its eventual impact, giving six distinct points.

(6 marks)

PART B (20 MARKS)

QUESTION 1

a) Discuss how morals, ethics and laws differ as guides for behaviour, and explain how the ACM code of ethics and professional conduct should shape a computing professional's actions.

(5 marks)

b) Explain five ways the ACM code of ethics and professional conduct should guide the software engineer's actions in this situation.

(5 marks)

QUESTION 2

a) State FIVE (5) points about software insecurity and buffer overflows, covering their causes, how a stack buffer overflow hijacks execution, and techniques used to strengthen software security.

(5 marks)

b) Explain FIVE (5) points about the Windows security mechanisms, covering the directory service used to organise resources, the registry hives holding security data, how access rights are encoded, and why the complexity of these mechanisms can itself be a weakness.

(5 marks)

END OF QUESTION PAPER

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